

ChatGPTChatGPT

### 3: Human Expertise-Work mode

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Continuing from 3: Human Agency and Expertise: Create a spreadsheet coding the entire human–AI interaction log for paper1-Log of human-AI collaborations.pdf. Extract every human prompt/turn in sequence and classify each using the proposed scheme: interaction types (Delegation, Directed delegation, Evaluation, Correction/Challenge, Redirection, Knowledge injection, Selection/Rejection, Deliberation), agency mode (Delegative, Directive, Evaluative, Critical, Deliberative), expertise evidence (e.g., domain correction, conceptual discrimination, evidence judgment, methodological judgment, argumentative judgment, genre/writing expertise, none), and outcome/notes. Preserve prompt order and include concise prompt excerpts plus page/line provenance where possible. Deliver an Excel-ready table/workbook and summarize key coding patterns.

Worked for 7m 37s >

Viewed 8 images >

Created the workbook covering **all 22 human turns**, in order, with excerpts, full prompts, interaction codes, agency modes, expertise evidence, rationale, outcomes, and provenance.

:codex-file-citation(path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/paper1-coding/paper1\_human\_AI\_coding.xlsx" purpose="output")

Also available: Excel-ready CSV and numbered source text.

Key patterns:

- **Directive agency dominates:** 10 turns; Critical 5; Delegative 3; Evaluative 3; Deliberative 1.
- Drafting and scrutiny alternate throughout. The sequence does not show a simple progression toward deliberation.
- Human interventions refine technical descriptions, reject wording, resolve terminology, and narrow evidence criteria.
- The exact-quotation request exposes the assistant’s unsupported Peleska claim.

This is **interpretive first-pass coding**, without independent coder validation. Line references identify extracted-text lines; the PDF has no printed line numbers. Turn order, full prompts, and summary counts were verified in the saved workbook.

- :codex-followup[Review ambiguous codes]{prompt="Review the ambiguous coding decisions in the Paper 1 workbook and propose justified alternatives."}
- :codex-followup[Draft the methods section]{prompt="Draft a coding-methods section based on the Paper 1 workbook, including operational definitions and limitations."}
- :codex-followup[Write the case findings]{prompt="Write a source-grounded Paper 1 findings section using the coded turns and representative excerpts."}

Edited 6 files+120-1

tmp/code\_log.py+48-0 tmp/build.mjs+67-0 tmp/build.mjs+1-0 Show 3 more files ▼

Do the same analysis for paper 2. Paper 2 has much more prompts. Paper 1 unfortunately I've lost most of the prompts.

Worked for 17s >

I can apply the same scheme, but **Paper 2’s interaction log isn’t in the files currently available to this chat**. Please attach its PDF here, including all parts if it spans multiple files.

I’ll code every human turn in sequence and deliver the same Excel workbook and CSV, with excerpts, provenance, coding rationale, outcomes, and a pattern summary.

Your clarification also changes Paper 1’s interpretation: it contains **22 surviving turns**, not the full collaboration. Comparisons should account for that missing-record limitation.

Worked for 14m 40s >

Viewed 8 images >

Completed the coding of **all 90 visible human turns** in Paper 2’s log.

:codex-file-citation(path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/paper2-coding/paper2\_human\_AI\_coding.xlsx" purpose="output")

The workbook includes full prompts, concise excerpts, all requested classifications, coding rationale, outcomes, page/line provenance, and summaries. Also available: Excel-ready CSV and numbered source text.

| Dominant agency mode | Turns | Share |
|----------------------|-------|-------|
| Directive            | 47    | 52.2% |
| Critical             | 19    | 21.1% |
| Evaluative           | 15    | 16.7% |
| Deliberative         | 6     | 6.7%  |
| Delegative           | 3     | 3.3%  |



The main pattern is **delegated drafting combined with repeated human control over evidence and framing**: correcting citations, challenging fabricated quotations, reducing the source pool, merging overlapping categories, and refining scope and genre.

Twelve turns have no substantive assistant reply visible; these outcomes are flagged. Repeated requests remain separate. This is interpretive first-pass coding, without independent coder validation. Paper 1’s missing prompts limit comparisons between the cases.

- :codex-followup[Review ambiguous coding][prompt="Review the ambiguous coding decisions in the Paper 2 workbook and propose justified alternatives."]
- :codex-followup[Write Paper 2 findings][prompt="Draft a findings section grounded in the Paper 2 coded turns and representative excerpts."]
- :codex-followup[Compare the two cases][prompt="Compare Paper 1 and Paper 2 coding patterns while explicitly accounting for Paper 1’s missing prompts."]

Edited 11 files+320-10

tmp/extract\_p2.py+12-0tmp/extract\_p2.py+1-1tmp/paper2/segment.py+7-0Show 8 more files

what does “**Expertise: none**” column mean?

“**Expertise: none**” means the turn provides insufficient visible evidence of expertise under the coding scheme. It does **not** mean the researcher lacks expertise.

For example, “yes” or “write Section 1” reveals little about domain knowledge, evidence judgment, or methodological reasoning—even when an expert writes it.

The column is a binary indicator:

- **1:** No specific expertise evidence was coded for that turn.
- **0:** At least one expertise category was coded.

A clearer label would be “**No explicit expertise evidence in this turn.**”

User attachment  
I also put the data as a stacked chart like this. Yeah, essentially the expertise as well, I guess. I added expertise as well. Shouldn't we add it like that?

Yes—**expertise evidence should be included in the visualization**, because it helps show where human steering draws on identifiable judgment. Your chart usefully preserves the sequence of turns.

I would, however, show expertise **in a separate, aligned panel**, rather than stack it on top of interaction types. They measure different things:

- **Interaction types:** what the human does—delegates, challenges, redirects, selects.
- **Expertise evidence:** what judgment the turn reveals—conceptual, evidential, methodological, argumentative, or editorial.

Combining them makes the total height difficult to interpret. A height of eight means eight assigned labels, **not greater expertise or stronger agency**. A short correction can be intellectually consequential despite receiving few labels.

For the paper, I suggest two panels sharing the same horizontal axis of **human turn number**:

1. **Interaction types present in each turn.**
2. **Expertise evidence present in each turn**, showing the individual categories rather than one aggregate “Expertise” layer.

A heatmap—with categories as rows and turns as columns—would make both panels clearer than stacked areas. Each marked cell would mean that a category is present, without implying continuity between discrete turns.

Exclude “**Expertise: none**” from any positive expertise total. If you retain an aggregate indicator, label it “**Number of expertise-evidence categories coded**”, rather than simply “Expertise.”

User attachment  
User attachment  
You mean like this, right? Two separate charts.

Yes—that is the separation I meant. The first chart shows interaction types; the second shows expertise evidence expressed in those same turns.

For the paper, make four small adjustments:

- Stack the charts vertically and align their plotting areas so each turn lines up.
- Label the horizontal axis “**Human turn number.**”
- Label the vertical axes “**Number of interaction types coded**” and “**Number of expertise-evidence categories coded,**” using whole-number ticks.
- Prefer **stacked columns** to stacked areas, because each turn is a discrete observation.

Keep “Expertise: none” excluded, as you have done. A zero in the second chart means **no explicit expertise evidence was coded in that turn**, not that expertise was absent.

The caption should also clarify that **height represents the number of assigned categories—not the strength of agency or level of expertise**.

User attachment  
User attachment  
good?

Yes—**these are much clearer**. The discrete columns, separate dimensions, and axis labels now represent the coding appropriately.

Only two finishing adjustments:

- Align the left and right edges of the plotting areas when placing the charts together, so corresponding turns line up.
- Add panel labels: **(a) Interaction types** and **(b) Expertise evidence**.

In the caption, state: “**Multiple categories may be coded within a turn. Column height indicates the number of categories coded, not the strength of agency or level of expertise. Zero expertise-evidence categories means no explicit evidence was identified in that turn.**”

"Domain correction" has no occurrences in this case; retaining it in the legend is useful if you use the same categories across all papers.

So I am adding data for another paper, number seven. For paper writing, I have two consecutive logs, one and two. Also, paper was asked to be revised during peer review, so that's also another log. So analyze all of them and generate the Excel file.

Worked for 3m 55s >

 Viewed an image >

 Edited 4 files+72-0

tmp/extract\_p7.py+20-0

tmp/paper7/diagnose.py+11-0

tmp/paper7/ocr.ps1+33-0

Show 1 more file ▼

I attached the logs again. I think this one should be in a readable format for you.

Worked for 15m 56s >

 Viewed 9 images >

`.codex-file-citation{path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/paper7-coding/paper7_human_AI_coding.xlsx" purpose="output"} covering all 132 visible human turns: 65 + 49 writing turns, followed by 18 revision turns.`

The workbook includes full prompts, excerpts, classifications, rationales, outcomes, page/line provenance, a codebook and chart-ready binary columns.

Key patterns:

- **Directive agency dominates:** 90 turns (68.2%).
- **Critical scrutiny:** 21 turns, particularly source attribution, terminology and revision arguments.
- **Expertise evidence:** visible in 92 turns, most often writing, methodological and argumentative judgment.
- Revision includes substantive human correction of the AI's interpretation of reviewer comments.

This is first-pass interpretive coding. "None" means no explicit expertise evidence in that turn. Unverified AI completion claims are flagged.

- `.codex-followup[Create the two charts]{prompt="Add separate stacked column charts for interaction types and expertise evidence to the Paper 7 workbook."}`
- `.codex-followup[Compare the papers]{prompt="Compare Papers 1, 2 and 7 using normalized coding patterns, accounting for Paper 1's missing prompts."}`
- `.codex-followup[Draft the findings]{prompt="Draft a research-paper findings section based on the Paper 7 coding, with representative prompt excerpts and provenance."}`

 Edited 7 files+405-6

tmp/paper7/segment.py+25-0

tmp/paper7/segment.py+1-1

tmp/paper7/specs.txt+132-0

Show 4 more files ▼

Do the same analysis for Paper 4. I attached all its files. I think it's quite comprehensive, right? That particular paper's interaction set.

Worked for 19m 56s >

 Viewed 8 images >

Yes—Paper 4 is comprehensive across **SLR planning, execution, and manuscript writing**.

`.codex-file-citation{path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/paper4-coding/paper4_human_AI_coding.xlsx" purpose="output"}`

The workbook includes **130 distinct human turns**, full prompts, classifications, coding rationales, outcome notes, page/line provenance, summaries, and a codebook. A separate sheet preserves all **136 exported occurrences**, including six shared-history duplicates.

Key patterns:

- **Directive agency dominates:** 84 turns (64.6%).
- Human oversight repeatedly addresses study selection, missing papers, extraction fidelity, and unsupported claims.
- Expertise evidence appears in **111 turns (85.4%)**, especially writing, methodological judgment, and evidence judgment.

Three AI replies are missing. I also flagged unsupported AI-generated counts. This is a first-pass coding of the available record; category counts describe observable behavior, not researcher competence.

- `.codex-followup[Create the two separate charts]{prompt="Create separate stacked column charts for Paper 4 interaction types and expertise evidence using the completed workbook."}`
- `.codex-followup[Compare Papers 2, 4, and 7]{prompt="Compare coding patterns across Papers 2, 4, and 7, accounting for different research phases and log coverage."}`
- `.codex-followup[Draft the Paper 4 findings]{prompt="Draft an empirical findings subsection for Paper 4 using the workbook, with concrete turn examples and appropriate limitations."}`

 Edited 9 files+384-4

tmp/extract\_p4.py+18-0

tmp/paper4/segment.py+39-0

tmp/paper4/segment.py+2-2

Show 6 more files ▼

By the way, paper what I gave you was about this topic: paper4-Factors quality of AI-generated code-a SLR. But inside the Excel file you have put the following, the following heading **"Paper 4: Human agency and expertise"**. Isn't that confusing? did you analyze the paper 4 data that I gave you?

Yes—I analyzed the **Paper 4 files you provided**: the SLR planning log, SLR execution log, and both paper-writing logs, using *Factors Influencing the Quality of AI-Generated Code: A Synthesis of Empirical Evidence* as context.

"Paper 4: Human agency and expertise" describes the **analysis being performed on those logs**, but it makes the source paper's identity unclear. You're right—the heading should instead be:

**Paper 4 — Factors Influencing the Quality of AI-Generated Code**  
*Human–AI interaction coding: agency and expertise*

The heading was ambiguous; the analysis used the correct Paper 4 dataset.

So you have analyzed the logs of paper 1, 2, 4, and 7. Can you put the number of turns or prompts? I don't like human turns. You can call human prompts in each of them and put them in a table?

Worked for 8s >

So you have analyzed the logs of paper 1, 2, 4, and 7. Can you put the number of turns or prompts?

| Paper   | Human prompts analyzed |
|---------|------------------------|
| Paper 1 | 22                     |
| Paper 2 | 90                     |
| Paper 4 | 130                    |
| Paper 7 | 132                    |
| Total   | 374                    |



Paper 4 contains 136 exported prompt occurrences; six duplicated exchanges were counted once. Paper 1's count covers only the surviving prompts, since much of its interaction history was lost.



So I'm giving you the list of the papers that I want to analyze. As you can see, there are six of them. I removed one of them because we didn't have, we had low number of prompts. I think we had 22. So that one is gone. Just checking your memory. And I also give kind of the topic name so that you know which file is which. I mean the numbers could have changed, but let's refer to papers, yeah, with their topic as well.

Yes—I understand the revised six-paper set. **The former Paper 1 (MBT), with 22 surviving human prompts, is excluded.** I'll identify each case by its current ID and topic, since the numbering has changed.

| Current ID | Topic                                                            | Earlier identification    |
|------------|------------------------------------------------------------------|---------------------------|
| P2         | Human oversight and overload in AI-assisted software engineering | Paper 2                   |
| P3         | SLR of AI-assisted coding                                        | Former Paper 4            |
| P4         | MLR of benefits and criticisms of ISTQB certifications           | ISTQB files just supplied |
| P5         | Reunification viability in divided ethnic populations            | Former Paper 3            |
| P6         | Diagnosing quality of higher education                           | Former Paper 5            |
| P7         | Characterizing human–AI interactions in co-writing               | Paper 7                   |



I also see **215 prompts for P2** in your table, whereas our earlier workbook contained **90**. I'll keep that discrepancy explicit until we reconcile the underlying logs. The table lists **296 for P5** and **122 for P6**; P3, P4, and P7 remain blank.

Okay, and, well, give me the count, right? So the number of prompts. I need to know how many number of prompts we have in the other ones. I just want to make sure if you have the log files or I need to give it to you.

Worked for 28s >

**You don't need to resend the ISTQB, reunification, or higher-education logs—I can access them in your folder.** Here is the current count status using your revised numbering:

| ID | Topic                                                            | Human prompts                 | Status                                                         |
|----|------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------|
| P2 | Human oversight and overload in AI-assisted software engineering | <b>90 previously analyzed</b> | Current log accessible; your table's <b>215</b> needs checking |
| P3 | SLR of AI-assisted coding                                        | <b>130</b>                    | Verified; excludes six duplicated exchanges                    |
| P4 | ISTQB certification benefits and criticisms                      | <b>Not yet counted</b>        | All three logs accessible                                      |
| P5 | Reunification viability                                          | <b>296 in your table</b>      | Log accessible; count not independently verified               |
| P6 | Diagnosing higher-education quality                              | <b>122 in your table</b>      | Log accessible; count not independently verified               |
| P7 | Human–AI interactions in co-writing                              | <b>132</b>                    | Verified; extracted logs and workbook retained                 |



The outstanding work is to count **P4** and verify **P2, P5, and P6** against the current files.

Well, I want you to verify the numbers again. So if needed, so verify all the files, count them. We need to check.

Worked for 3m 30s >

 Viewed 2 images >

 Edited 2 files+22-1

tmp/recount.py+18-0tmp/recount.py+4-1

only analyze the PDF files in "Data (empirical)" folder

Worked for 20s >

only analyze the PDF files in "Data (empirical)" folder . In the PDF file names, I don't use the paper numbers anymore, but you have the title and you can look inside them.

Worked for 1m 17s >

I recounted the readable logs **directly in "Data (empirical)"**, using their topics rather than old paper numbers.

| Paper topic                                                      | Verified human prompts               |
|------------------------------------------------------------------|--------------------------------------|
| Human oversight and overload in AI-assisted software engineering | 90                                   |
| SLR of AI-assisted coding                                        | 130                                  |
| ISTQB certification benefits and criticisms                      | 132                                  |
| Reunification viability                                          | 30                                   |
| Diagnosing quality of higher education                           | Cannot yet verify—file access denied |
| Human–AI interactions in co-writing                              | Log absent from this folder          |
| Verified subtotal: four cases                                    | 382                                  |



The SLR has **136 exported occurrences**, including six duplicated exchanges counted once. ISTQB has **65 + 49 + 18 = 132** prompts across its three logs.

**I previously misidentified the ISTQB workbook as the co-writing case.** The 132 count belongs to ISTQB. The reunification log explicitly continues an earlier chat, so its 30 prompts cover only the supplied continuation.

To finish checking all six, I need a readable copy of the higher-education log and the co-writing log placed in this folder. The earlier figures **215, 296, and 122 remain unverified**.

oSaved count-verification notes

 Edited 2 files+43-0

tmp/count\_current.py+18-0tmp/verify\_counts.py+25-0

See these files attached and count them.

Worked for 4m 8s >

 Viewed 5 images >

| Paper topic                         | Human prompts in the supplied log | PDF pages |
|-------------------------------------|-----------------------------------|-----------|
| Reunification viability             | 30                                | 37        |
| Human–AI interactions in co-writing | 77                                | 127       |
| Total                               | 107                               | 164       |



The reunification file is unchanged and explicitly continues an earlier chat; **30 covers this supplied log only**.

For the co-writing log, I counted human-message boundaries and joined prompts continuing across pages. The manuscript PDFs provide context and do not add prompts.

 Edited 2 files+20-0

tmp/count\_current.py+1-0tmp/count\_blue.py+19-0

See the attached two files. I guess you have misunderstood or you have mistaken those. You said you had problems with the PDF, so I converted the PDF to Word files. And I don't want you to tell me how many pages are the PDFs.

Worked for 3m 38s >

I checked both Word files, excluding conversion duplicates and joining prompts split across text boxes.

| Paper topic                            | Human prompts |
|----------------------------------------|---------------|
| Diagnosing quality of higher education | 122           |

| Paper topic             | Human prompts |
|-------------------------|---------------|
| Reunification viability | 30            |
| Total                   | 152           |



The higher-education count **confirms your 122 figure**. The reunification Word file confirms **30 prompts in the supplied log**, which begins as a continuation of an earlier chat.

Edited 6 files+63-1

tmp/word\_count.py+13-0 tmp/word\_diag.py+12-0 tmp/word\_boxes.py+13-0 Show 3 more files

:codex-annotation(index="1") I counted the entries in the list you pasted: it contains **30 human prompts**, from "So I continue my work here..." to "trans to AZ, TURKCE and FARSI..."

The pasted list therefore supports **30**, rather than 57. Does **57** include an earlier reunification chat, or both human prompts and AI replies?

there are 57 lines in its TOC:

Worked for 11s >

Remember you had analyzed the log files and generated this kind of coded Excel files for several of the papers. I attached one of them. You can also review what you did. I want you do that for the following two papers also and gen a XLS file for each



Diagnosing quality of higher education



Characterizing human-AI interactions in co-writing

Worked for 22m 5s >

Viewed 12 images >

Created both workbooks using the structure of your example:

- Diagnosing quality of higher education — 119 human prompts:** :codex-file-citation(path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/two-case-coding/Diagnosing\_higher\_education\_coded\_interactions.xlsx" purpose="output")
- Characterizing human-AI interactions in co-writing — 82 human prompts:** :codex-file-citation(path="C:/Users/3054738/.codex/.chatgpt-projects/g-p-6ab3efeb32ec819194c3f2e17f283c89/outputs/two-case-coding/Human\_AI\_co\_writing\_coded\_interactions.xlsx" purpose="output")

Each contains **Summary, Coding, Human prompts, and Codebook**, including full prompts, excerpts, interaction types, agency modes, expertise evidence, rationales, outcomes, provenance, and chart-ready columns.

Both show frequent directed drafting and critical scrutiny. Higher education emphasizes causal-model development and evidence relevance; co-writing includes more deliberation about conceptual framing and methodological choices.

The corrected counts replace **122 and 77**, respectively. Higher education uses the converted Word log; co-writing required OCR, so quotation wording should be checked against the PDF. Workbook totals and formulas were verified.

- :codex-followup[Add the two stacked charts]{prompt="Add separate stacked column charts for interaction types and expertise evidence to both workbooks."}
- :codex-followup[Compare all paper cases]{prompt="Create a comparison workbook using all completed coded workbooks, with counts and normalized percentages by paper topic."}
- :codex-followup[Review borderline codes]{prompt="Review the ambiguous and borderline classifications in these two workbooks and provide proposed coding refinements with supporting prompt excerpts."}

Edited 17 files+484-13

tmp/new\_cases\_diag.py+13-0 tmp/new\_cases\_diag.py+2-2 tmp/render\_cowriting.py+8-0 Show 14 more files

how to differentiate between

Delegation and

Directed delegation?



The distinction is **how much substantive direction the human supplies for the AI's task**.

| Type       | Coding rule                                                                          | Example                   |
|------------|--------------------------------------------------------------------------------------|---------------------------|
| Delegation | Requests an output without specifying substantive content, approach, or constraints. | "Write the introduction." |

| Type                       | Coding rule                                                                                       | Example                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Directed delegation</b> | Specifies what the output should contain, how it should be produced, or criteria it must satisfy. | "Write the introduction, emphasize the causal model's contribution, avoid policy recommendations, and use verified references." |



A useful question is: **Does the prompt leave the main execution decisions to AI, or does the researcher specify those decisions?**

For consistent coding, simply naming a section—"Give Section 5.2"—should normally count as **Delegation**. Naming a section alone does not demonstrate substantive direction. If the prompt explicitly refers to an agreed plan—"Write Section 5.2 using the three mechanisms we agreed"—it can count as **Directed delegation**.

**Some rows in the workbooks treated named-section requests as Directed delegation because of earlier context. That boundary needs standardizing** using the rule above. Neither category, by itself, establishes the researcher's expertise.

Suggest revision to delegation phrase to ensure they are clearly distinct.

I suggest **Open delegation** and **Directed delegation**:

| Revised label              | Definition                                                                                                                              |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Open delegation</b>     | Requests AI to generate, revise, or explain something <b>without specifying substantive content, approach, or evaluation criteria</b> . |
| <b>Directed delegation</b> | Requests AI to perform a task <b>with explicit substantive instructions about content, approach, scope, or evaluation criteria</b> .    |



Examples: "Write Section 3" = **Open delegation**; "Write Section 3, explain the survey design, and distinguish observations from empirical evidence" = **Directed delegation**.

"Open" refers to the discretion granted to AI in that prompt; earlier conversation context may still constrain the task.

Also we need to differentiate between knowledge injection and expertise. Expertise has several subtypes.

Yes. **Knowledge injection describes what the human supplies; expertise evidence describes the judgment the human demonstrates**. They belong to separate coding dimensions.

For clarity, I suggest renaming **Knowledge injection** to **Knowledge/context provision**.

| Dimension                                             | Definition                                                                                                   | Example                                                                                              |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Knowledge/context provision</b> — interaction type | Supplies information, sources, drafts, data, experience, or interpretations for AI to use.                   | "Here is the survey dataset; use it in Section 4."                                                   |
| <b>Expertise evidence</b> — separate dimension        | Demonstrates a substantive judgment about concepts, methods, evidence, arguments, domain claims, or writing. | "These survey responses indicate perceived quality; they do not establish actual learning outcomes." |



The expertise subtypes remain **domain correction, conceptual discrimination, evidence judgment, methodological judgment, argumentative judgment, and genre/writing expertise**.

The two dimensions can occur independently or together:

| Human prompt                                                                                      | Knowledge/context provision           | Expertise evidence                                |
|---------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------|
| "Here is the report."                                                                             | Yes                                   | None observable                                   |
| "This report concerns primary schools, so it cannot directly support our higher education claim." | Yes                                   | Evidence judgment; conceptual discrimination      |
| "Your causal-loop label is incorrect because our model contains no feedback loops."               | Yes—supplies a corrective explanation | Domain correction; conceptual discrimination      |
| "Shorten this paragraph to 100 words."                                                            | No new knowledge/context              | Genre/writing expertise, under the current scheme |



The key coding rule is: **Supplying material alone does not demonstrate expertise. Code expertise only when the prompt makes the relevant judgment observable**. "None" means no explicit evidence in that prompt, not that the researcher lacks expertise.

better phrase for Selection/Rejection?

I suggest **Acceptance/Rejection**.

**Definition:** Explicitly accepts, selects, or rejects an AI-proposed option, output, recommendation, or next step.

Examples:

- "Use the second title." → Acceptance
- "Yes, proceed with that outline." → Acceptance
- "Remove this suggestion; it is outside our scope." → Rejection

This captures the **decision made**. **Deliberation** captures reasoning about alternatives, while **Redirection** captures changing the course of work. A prompt can receive more than one code when it performs several acts.

What we should include as the group phrase for all these? You call them expertise evidence. Should we call them types of expertise by human in the prompt?

| Expertise evidence        |
|---------------------------|
| domain correction         |
| conceptual discrimination |
| evidence judgment         |
| methodological judgment   |
| argumentative judgment    |
| genre/writing expertise   |



I recommend **"Expertise demonstrated in the prompt"** as the group heading. For a shorter chart label, use **"Demonstrated expertise."**

"Types of human expertise" could imply that we are classifying the researcher's overall abilities. Your analysis identifies expertise **expressed through particular judgments in each prompt**.

To make the subtypes consistent, use:

| Demonstrated expertise   |
|--------------------------|
| Domain-specific judgment |
| Conceptual judgment      |
| Evidence judgment        |
| Methodological judgment  |
| Argumentative judgment   |
| Genre/writing judgment   |



Define the dimension as: **"Observable human judgments in a prompt that draw on domain, conceptual, evidential, methodological, argumentative, or writing knowledge."**

Changing **domain correction** to **domain-specific judgment** broadens that category: expertise can appear through a technically informed explanation or qualification, as well as a correction. Existing codes would therefore need reviewing against the revised definition.

For "none," use **"No explicit expertise demonstrated."**

Should we call it perceived human expertise? Perceived means by looking at, by observing and analyzing after the fact.

**"Perceived human expertise" is defensible**, but "perceived" may suggest someone's impression of the researcher's competence rather than an interpretation of the prompts.

I recommend **"Observed indicators of human expertise."** This makes clear that you analyze visible expressions of expertise without claiming to measure the researcher's overall competence.

You could define it as:

Observed indicators of human expertise are prompt features interpreted through retrospective analysis as expressing domain-specific, conceptual, evidential, methodological, argumentative, or genre/writing judgment.

For a shorter table or chart heading, use **"Observable expertise indicators."** For "none," use **"No explicit indicator observed."**

what did you mean by Deliberation  
in prompt (Interaction) type? a better phrase for it?

By **Deliberation**, I meant that the human **weighs alternatives, examines trade-offs, or reconciles competing ideas before deciding how to proceed**.

I suggest “**Reasoned comparison**” as a clearer interaction-type label.

| Example prompt                                                                                                                           | Classification                                            |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| “Should we strengthen the empirical analysis or narrow our claims? Stronger analysis would help, but the available logs are incomplete.” | <b>Reasoned comparison</b>                                |
| “Give me three possible approaches.”                                                                                                     | Delegation—it requests alternatives without weighing them |
| “Use the second approach.”                                                                                                               | Acceptance/Rejection—it states a choice                   |



**Definition:** “Explicitly compares alternative concepts, methods, interpretations, or writing choices by considering their merits, limitations, or consequences.”

This captures the reasoning process; **Evaluation** assesses an idea or output, while **Acceptance/Rejection** records the decision. A prompt may do more than one of these.

 User attachment

 User attachment

These two are two images that I've generated from the Excel file of P1. Give a proper— I'm gonna put them together. Give a proper figure caption.

RE: , rather than the strength of agency or expertise. --- I'm going to remove that phrase because the strengths of agency or expertise in each prompt. I think it depends. It depends if the researcher felt the need to inject his expertise. We can't expect each prompt to have high level of expertise, right?

Yes—you cannot expect every prompt to express expertise. A short request may simply advance an already agreed task. **The absence of an expertise code does not indicate low researcher expertise.** Whether the researcher felt intervention was needed is a plausible explanation, but the prompt alone cannot establish that motivation.

You can remove the phrase. Keep the statement defining bar height as the **number of categories coded**, since these charts do not directly measure strength.

I think the last sentence you should say that usually or generally it means that the researcher did not feel the need to inject it.

We can include that interpretation, but “generally means” would imply that the logs establish the researcher’s motivation. **“May reflect”** preserves your point more accurately.

And the following is a table that I've taken from your Excel file. Give a caption for table. Keep it very short because we will discuss it in text. Table 4

|                                      |                                    |            |              |
|--------------------------------------|------------------------------------|------------|--------------|
| Phase of co-writing                  | First turn                         | Last turn  | # of prompts |
| (turns)                              | Classification of prompts by human |            |              |
| Delegative                           | Directive                          | Evaluative | Critical     |
| Deliberative                         |                                    |            |              |
| Initial framing                      | 1                                  | 7          | 7            |
| 1                                    | 4                                  | 1          | 1            |
| 0                                    |                                    |            |              |
| Source selection and citation repair | 8                                  | 19         | 12           |
| 1                                    | 7                                  | 0          | 4            |
| 0                                    |                                    |            |              |
| Abstract and paper positioning       | 20                                 | 36         | 17           |
| 0                                    | 7                                  | 5          | 3            |
| 2                                    |                                    |            |              |
| Conceptual structure                 | 37                                 | 44         | 8            |
| 0                                    | 6                                  | 0          | 0            |
| 2                                    |                                    |            |              |
| Supplement and section drafting      | 45                                 | 59         | 15           |
| 0                                    | 10                                 | 1          | 4            |
| 0                                    |                                    |            |              |
| Evidence audit and pool revision     | 60                                 | 69         | 10           |
| 0                                    | 3                                  | 4          | 3            |
| 0                                    |                                    |            |              |
| Completion and review requests       | 70                                 | 78         | 9            |
| 1                                    | 5                                  | 2          | 1            |
| 0                                    |                                    |            |              |
| Feedback and reframing               | 79                                 | 90         | 12           |
| 0                                    | 5                                  | 2          | 3            |
| 2                                    |                                    |            |              |

I checked the literature and agency modes are the followings. Why aren't we using them? Human agency operates through three primary modes—**individual (personal), proxy, and collective**—as defined by psychologist Albert Bandura in social cognitive theory.

Worked for 20s >

You are right about Bandura’s three modes. **I should not have labelled our five categories “agency modes” without distinguishing them from that established framework.** They were proposed coding categories for this study, not Bandura’s categories.

Bandura distinguishes **personal agency**—acting directly; **proxy agency**—relying on others to act on one’s behalf; and **collective agency**—acting through coordinated, interdependent effort. [Bandura \(2001\)](#)

The two classifications address different questions:

| Classification                | Question it addresses                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bandura’s agency modes</b> | Through whom, or through what social arrangement, is agency exercised?                                                                              |
| <b>Our prompt categories</b>  | How does the researcher intervene in this particular exchange—by delegating, directing, evaluating, challenging, or reasoning through alternatives? |



For example, “Draft this section using my argument and these sources” combines **personal direction** with delegated execution. Interpreting AI execution as **proxy agency** would be an adaptation of Bandura’s concept, which refers to others acting on one’s behalf. Similarly, human–AI collaboration alone does not establish collective agency in Bandura’s sense.

**My recommendation is to use Bandura as a theoretical foundation and rename our column “Form of human intervention.”** The five labels can remain **Delegative, Directive, Evaluative, Critical, and Deliberative**, explicitly described as study-specific operational categories.

Table 4’s caption would then become:

**Table 4. Distribution of human prompts by form of human intervention across co-writing phases for P1.**

We should also examine whether this five-category dimension adds enough information beyond the interaction types: delegation/delegative and evaluation/evaluative currently overlap considerably. Renaming resolves the terminology problem; clarifying their distinct analytical purpose resolves the methodological one.

Intervention could become more of a like a harsh phrase. Can you suggest other alternatives?

I recommend “**Human engagement approach**”. It is neutral and describes how the researcher engages with AI in each prompt.

Other options:

| Group heading               | Emphasis                                        |
|-----------------------------|-------------------------------------------------|
| Human engagement approach   | How the researcher engages with AI              |
| Form of human participation | How the researcher participates in the exchange |
| Human guidance approach     | How the researcher guides AI’s work             |
| Human agency expression     | How agency is expressed in the prompt           |



“**Human agency expression**” is strongest if you want to retain the connection to agency theory while distinguishing your categories from Bandura’s modes.

For Table 4: “**Distribution of human prompts by agency expression across co-writing phases for P1.**”

And the other one, which is expertise, is nice. So expertise and guidance, do they go nice together? I like guidance, but I’m not sure if guidance and expertise go together, because that’s gonna also go to the paper title.

Yes—**human guidance and expertise** work well together. Guidance describes **how the researcher shapes AI’s work**; expertise describes **the knowledge and judgment expressed in doing so**.

They are related but distinct: a researcher can give a simple instruction without expressing substantive expertise, or use expertise to challenge an interpretation and guide the next draft.

For your paper, I suggest:

**From Delegation to Deliberation: Human Guidance and Expertise in AI-Assisted Academic Writing**

One qualification: **guidance** emphasizes steering the collaboration, whereas **agency** also encompasses autonomy, choice, and intellectual ownership. If the paper primarily analyzes prompts and their observable functions, “guidance” fits well. The coding dimension could be called “**Forms of human guidance**”, alongside “**Observable expertise indicators**.”

I like the word agency, but I think I will keep that for a future paper. We can mention a future work, because I think that needs more work. That’s my gut instinct. We can just keep guidance and expertise.

Yes. **Guidance and expertise** give this paper a focused scope grounded in the prompts you can observe. A fuller analysis of agency can remain future work.

The title becomes:

**From Delegation to Deliberation: Human Guidance and Expertise in AI-Assisted Academic Writing**

Use these headings consistently:

- **Interaction types** — what the human prompt does.
- **Forms of human guidance** — delegative, directive, evaluative, critical, and deliberative.
- **Observable expertise indicators** — the judgments expressed in the prompt.

For future work, you could write: “Future research could examine how these patterns of guidance and expertise relate to human agency, including autonomy, intellectual ownership, and control over consequential decisions.”

For the paper title, I have this phrasing in mind, and I had that. So I think we should change the human guidance. Should we say human guidance of AI or something? Examining Human Guidance and Expertise in AI-Assisted Academic Writing: A Retrospective Process Analysis

Yes—“**human guidance of AI**” makes the relationship clearer. I suggest:

**Examining Human Guidance of AI and Expressions of Expertise in Academic Writing: A Retrospective Process Analysis**

“Expressions of expertise” also matches what you analyze: expertise made observable in the prompts, rather than the researcher’s overall competence.

If you prefer to preserve your original phrasing more closely:

**Examining Human Guidance of AI and Expertise in AI-Assisted Academic Writing: A Retrospective Process Analysis**

The first version reads more smoothly and avoids repeating “AI.”

Why are we saying process analysis? We are not analyzing the processes. Is that precise? Should we just say retrospective analysis of logs or something?

You're right. **"Process analysis"** suggests examining stages, transitions, or how the collaboration develops. Coding prompts in sequence does not, by itself, establish that kind of analysis.

**"Retrospective analysis of interaction logs"** describes your work more precisely. I suggest:

**Examining Human Guidance and Expertise in AI-Assisted Academic Writing: A Retrospective Analysis of Human–AI Interaction Logs**

This retains your preferred concepts and clearly identifies the data being analyzed. In the methods, we can specify that "expertise" refers to observable indicators in the prompts.

Suggest better phrases for the followings.

**Domain correction**

Conceptual discrimination



I suggest:

| Current label             | Revised label                     | Meaning                                                                       |
|---------------------------|-----------------------------------|-------------------------------------------------------------------------------|
| Domain correction         | <b>Domain-specific correction</b> | Corrects a technical or factual claim using subject knowledge.                |
| Conceptual discrimination | <b>Conceptual differentiation</b> | Distinguishes concepts, categories, or meanings that should not be conflated. |



These clarify the wording while preserving the existing coding definitions. **"Domain-specific judgment"** is broader, but would require reviewing the codes because it includes explanations and assessments beyond correction.

**Domain-specific correction** : subject-matter **correction** ?

Yes—**Subject-matter correction** is clear and natural.

**Definition:** Correcting a factual or technical claim using knowledge of the subject.

It pairs well with **Conceptual differentiation**: the first corrects a claim; the second distinguishes concepts or meanings.

Okay, now it's time to start writing the paper. Look at the draft structure and contents that I have and then I will tell you to start which part.

Worked for 21s >

I've read the draft, including its tables and figure captions. Its structure is:

1. Introduction
2. Background and related work
3. Framework: interaction types, human guidance, and expertise indicators
4. Illustrative analysis of four paper cases
5. Synthesis of findings
6. Disclosure of AI use

The four cases are now **P1 Human oversight (90 prompts)**, **P2 ISTQB (132)**, **P3 Higher education (119)**, and **P4 Human–AI co-writing (82)**.

A few points to address as we write:

- Earlier text still refers to **agency**, three cases, and old paper numbers; these need alignment with the current scope.
- The sample table contains **MBT/fault-tolerance prompts**, although current P1 is Human oversight. Those examples need replacing.
- The classification procedure is best described provisionally as **AI-assisted qualitative coding with researcher review**. The described procedure alone does not justify calling it grounded theory.
- The framework needs a clear distinction between **interaction types** and **forms of human guidance**, given their overlapping labels.
- The AI disclosure names GPT-5.6 Sol, while the framework-development note names GPT-6.1; these roles and model details need reconciling.

I haven't changed the document or started drafting. I'm ready for the section you choose.

User attachment

OCR

Levels and Analytical Perspectives in Human–AI Scholarly Work

From Research Problems to Co-Writing and Retrospective Analyses

User attachment

User attachment

let's start with 2.2. I have put Figure 1 and Table 1 there to differentiate between this work and my recent work (attached), okay? And we need to be very short. Use the, write very concisely because the space will be limited. So give me 2.2 and refer / disucss to the figure and table. ----- Figure 1- Levels and analytical perspectives in human–AI scholarly work: From research problems to co-writing and retrospective meta analysis of co-writing process using interaction logs. Also give a caption for Table 1 and change Table 1 if needed.

Worked for 54s >

The attached Word file remains unreadable despite access being granted. The text below follows your figure, comparison table, and earlier draft.

In Figure 1, change “**Agency-and-Expertise Analysis**” to “**Guidance-and-Expertise Analysis**” and update the current paper’s title. Use “**retrospective analysis**” rather than “meta-analysis,” which could suggest statistical synthesis across studies.

I phrased section 2.2 as follows : Context of this work. Isn't that good? Also you didn't refer to the table in the text.

Yes—“**Context of this work**” fits well because the section positions this study relative to your earlier work. The table reference should be more substantive.

Okay, what you have written is extremely short. I didn't say you should write that shortly.

I revised: ... focusing on how the researcher guides AI and where subject knowledge and scholarly judgment of the human researcher is provided to the AI in the prompts.

“**Are conveyed**” reads more naturally than “is provided” and agrees with the two subjects:

| ...focusing on how the researcher guides AI and how the researcher’s subject knowledge and scholarly judgment are conveyed to AI through the prompts.

At Level 1, the researcher and AI draw on these foundations to co-write a manuscript.

Yes—this is clear and precise:

| At Level 1, the researcher and AI draw on these foundations to co-write a manuscript.

RE: The resulting interaction logs provide material for retrospective analysis: Level 2A characterizes the interactions, whereas Level 2B examines human guidance and expertise expressed within them. The two perspectives therefore address different analytical questions using records of the same underlying activity.--- You need to say why analyzing those logs can be useful. So you need to motivate also all of that.

And then you say that you shouldn't be so deterministic. You should say that a human researcher may decide to get AI's assistant in writing the paper out of his research work.

This makes AI involvement optional and distinguishes the research from writing assistance. :codex-annotation{index="1"}

| At Level 1, a human researcher may choose to seek AI assistance in writing a manuscript based on their research.

proof read:L When human gets assistance from AI for co-writing a paper, one side data source of the process is the log of interactions, which preserve exchanges through which the manuscript was developed, including instructions, proposed revisions, corrections, and decisions that may be invisible in the final text.

When a human researcher uses AI assistance to co-write a paper, the interaction log provides an additional data source documenting the manuscript’s development. It preserves exchanges—including instructions, proposed revisions, corrections, and decisions—that may be invisible in the final text.

I think there was something called research journal or research diary, right? So in a loosely coupled way, I think this log is similar to that, right?

Yes—a **loose analogy to a research diary or reflective journal is appropriate**. Research diaries record decisions and reflections that can later help researchers examine how their work developed. [Engin \(2011\)](#)

The distinction is that a diary is deliberately written for documentation or reflection, while an interaction log records exchanges generated during the work. It captures only the reasoning communicated to AI, rather than everything the researcher thought or did.

For your paragraph:

give 3 FRAMEWORK

3.1 CLASSIFYING TYPES OF HUMAN PROMPTS (INTERACTION TYPES)

Say how we derived the classification. Approach was: author used the GPT-6.1 model asking the AI to analyze all human prompts in P1 and P2 (see section 4.1) and suggest a draft classification. He then reviewed and approved the classification. Is that “grounded theory”, like qualitative coding?

List the items, and give example prompts for each, in a table

- Open Delegation
- Directed delegation
- Evaluation
- Correction/Challenge
- Redirection
- Knowledge/context provision
- Acceptance/Rejection of AI output
- Deliberation

Worked for 13s >

give 3 FRAMEWORK

3.1 CLASSIFYING TYPES OF HUMAN PROMPTS (INTERACTION TYPES)

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List the items, and give example prompts for each, in a table

- Open Delegation
- Directed delegation
- Evaluation
- Correction/Challenge
- Redirection

- Knowledge/context provision
- Acceptance/Rejection of AI output
- Deliberation----- Also always feel free suggest better heading phrases.

I suggest “**Classifying human prompts by interaction function**” for Section 3.1: it makes the basis of classification explicit.

The procedure is best described as **AI-assisted, data-informed qualitative classification**. Deriving categories from prompts does not by itself constitute grounded theory, which involves a broader theory-building methodology, including constant comparison and theoretical sampling. [Chun Tie, Birks, and Francis \(2019\)](#)

RE:&#x20;

**Requests generation, revision, or explanation without specifying substantive content, approach, or assessment criteria. --- Very concisely and shortly you also need to add that it could be that the researcher did not feel the need to give extra information because from the existing context that has been already going on in the conversation or from previous drafts, the AI can give a proper introduction, for example.**



Requests generation, revision, or explanation without additional substantive instructions, potentially because the researcher considers the existing conversation or draft sufficient context.

RE: Open and directed delegation are distinguished by the substantive instructions expressed in the prompt. Merely naming a section does not establish directed delegation, whereas explicitly invoking an agreed plan can do so. Evaluation assesses adequacy; correction/challenge identifies a particular concern. Deliberation considers reasons for alternative courses, while acceptance/rejection records a choice. Knowledge/context provision does not automatically demonstrate expertise: the latter requires an observable judgment, as discussed in Section 3.2. --- Your tone of message here does not seem like academic papers. It seems like you are writing like legal documents. You need to change your tone to academic writing.

give 3.2 Classifying types of demonstrated human expertise and guidance of AI in interactions

Say how we derived the classification. Same approach as above.

List the items, and give example prompts for each, in a table  
expertise :

- Subject-matter correction
- Conceptual differentiation
- Evidence judgment
- Methodological judgment
- Argumentative judgment
- Genre/writing expertise
- None: keep this, since we also want to know which prompts did not include any human expertise

guidanc&#x65;&#x20;&#x6F;f AI:

- Delegative &#x20;
- Directive &#x20;
- Evaluative &#x20;
- Critical &#x20;
- Deliberative

...

Show more ▼

give 4 ILLUSTRATIVE ANALYSIS: A SELECTED SET OF HUMAN-AI CO-WRITTEN PAPERS

4.1 DATASET: FOUR PAPER CASES CO-WRITTEN BY HUMAN AND AI

As Table 5

Table 5-Four selected paper cases co-written by human and AI

Paper case Paper citation AI model used for co-writing # of prompts issued by researcher

ID Field Topic

P1 Software engineering Human oversight and overload in AI-assisted software engineering [1]

GPT-5.2 90

P2 MLR of benefits and criticisms of the ISTQB certifications [2]

GPT-4.1 132

P3 Higher education Diagnosing quality of higher education [3]

Claude Sonnet 5 119

P4 Technical writing Characterizing human-AI interactions in co-writing [4]

GPT-6 82

RE: Each prompt constitutes a coding unit --- revise, not clear

Each human prompt is analyzed individually and assigned codes for its interaction types, predominant form of guidance, and observable expertise indicators.

Also, they may say why did we selected these ones? Say that, I mean, they are not really easy to find publicly available data. So the researcher used his own recent papers. These were kind of the ones with quite high number of prompts. So this was like a random stratified sampling or something from out of about ten papers that he has used using AI.

This is better described as **purposive case selection from an accessible corpus**, rather than stratified random sampling: the cases were selected for their available logs and substantial prompt counts, rather than randomly drawn.

give 4.2 CODING OF HUMAN PROMPTS FOR EACH PAPER ----- you can find the images and tables in the Word file

4.2.1 P1 (Human oversight)

Figure 2

Figure 3

(a) Interaction types coded in each human prompt. (b) Expertise indicators coded in the corresponding prompt. Each coloured segment represents one category; stacked heights indicate the number of categories coded. Prompts may receive multiple codes. A zero in panel (b) indicates that no explicit expertise indicator was identified and may reflect that the researcher did not feel the need to contribute expertise explicitly at that stage.

Figure 2-Interaction types (top) and observed indicators of human expertise (bottom) across the prompt sequence for P1.

Vvv

P1 provides a concrete illustration of how such iterative co-writing develops across a sequence of exchanges. Figure 1 presents an excerpt from the archived interaction record as a sequence diagram. Sequence diagrams, commonly used in the Unified Modeling Language (UML), represent interactions between participants in temporal order [5]. Here, the notation is adapted to show exchanges between the human researcher and AI during the early development of P1.

Figure 3- Excerpt from the human–AI co-writing interaction sequence for P1, represented as a UML sequence diagram.

The interaction began with the researcher providing an existing MSc report and specifying that both the structure and writing of the resulting journal paper would need substantial redevelopment. AI first reviewed the material and summarized the underlying work. The researcher then requested alternative paper titles and an outline, evaluated the proposed structure, modified it independently, and returned the resulting outline to AI for critique. Subsequent exchanges moved from paper-level architecture to the structured abstract, terminology, keywords, introduction, and individual sections. The sequence therefore illustrates movement across writing levels and purposes rather than a single request followed by AI-generated manuscript text.

Table 6

...

Show more ▾ ion of human prompts by guidance modes across co-writing phases for P1.

Worked for 1m 43s > turn Last turn # of prompts

(turns): Frequency of human guidance modes observed in each phase

try again

Delegative Directive Evaluative Critical Deliberative

Worked for 46s > 1 4 1 1 0

Source selection and citation repair 8 19 12 1 7 0 4 0

Viewed 6 images >

Abstract and paper positioning 20 36 17 0 7 5 3 2

I can read the file now. **Figure 3 depicts the MBT paper, not P1 (Human oversight)**: it explicitly mentions the student report on model-based testing. The MSc-report narrative therefore needs removing from this subsection. The text below uses Figure 2, Table 6, and the verified P1 coding.

Conceptual structure 37 44 8 0 6 0 0 2

Supplement and section drafting 45 59 15 0 10 1 4 0

Give the following subsections for P2 I have provided the figure, but for the other, for P3 and P4, keep the discussion short. I'm not putting the figures to save space, but completion discussion highlights. Yes 70 78 9 1 5 0 0 0

Completion discussion highlights 70 78 9 1 5 0 0 0

Feedback and reframing 79 90 12 0 5 2 3 2

Figure 4

jkj

Figure 4-Interaction types (top) and observed indicators of human expertise (bottom) across the prompt sequence for P2.

4.2.3 P3 (Diagnosing quality of higher education)

4.2.4 P4 (Characterizing human-AI interactions)

Worked for 39s >

Good job in giving some examples, but give a bit more examples for, for example, for the expertise as well for P2. You say, you know, knowledge provision, and give real example messages from the Excel files because you have them. Give prompt examples.

Worked for 28s >

The revision below adds quotations from the coded P2 log, identified by prompt number. :codex-annotation{index="1"}

Give P3 and P4 subsections too.

Regenerate P1 subsection as well by adding some code prompt example ones.

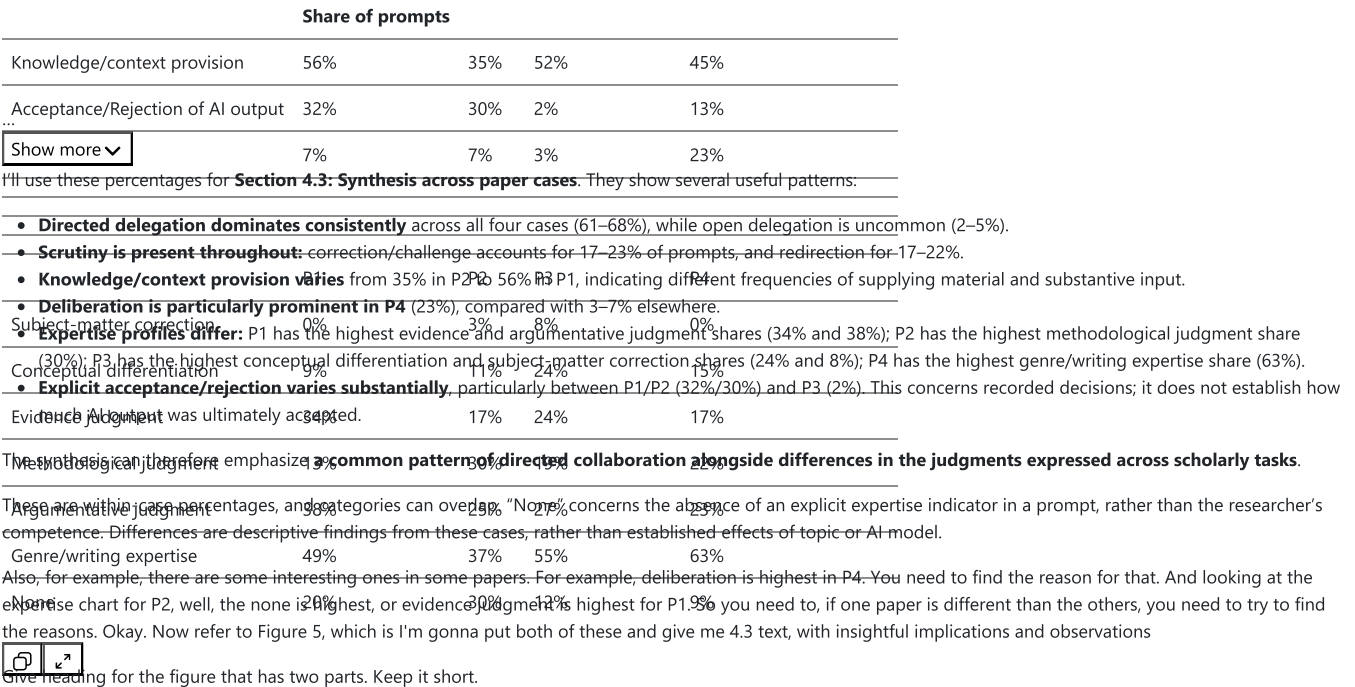
Worked for 22s >

User attachment

User attachment

I provide the data. Take this and understand them for the aggregator, for the section 4.3.&#x20;

|                      | Share of prompts |       |                  |                         |
|----------------------|------------------|-------|------------------|-------------------------|
|                      | Human oversight  | ISTQB | Higher education | Characterizing human-AI |
| Interaction type     | P1               | P2    | P3               | P4                      |
| Open Delegation      | 3%               | 5%    | 2%               | 2%                      |
| Directed delegation  | 64%              | 61%   | 68%              | 62%                     |
| Evaluation           | 21%              | 12%   | 19%              | 17%                     |
| Correction/Challenge | 21%              | 17%   | 23%              | 21%                     |
| Redirection          | 18%              | 17%   | 22%              | 20%                     |



**Figure 5. Cross-case comparison of (a) interaction types and (b) human expertise indicators.**

5 DISCUSSIONS AND IMPLICATIONS

And limitations

6 CONCLUSIONS

And future

give 5 DISCUSSIONS AND IMPLICATIONS

And limitations

6 CONCLUSIONS

And future ----- include these: say that the goal of this paper was to characterize, really to shed some light on this new emerging phenomenon of human-AI co-writing of technical writing and papers, which although many members of the research community and authors see it as like a taboo, and many journals want to reduce that, but it's just happening, and, like, too much resistance is not going to work. We need to understand and characterize and find best practices for proper approaches for this co-writing. This paper is one effort in that direction.

Worked for 1m 3s >

Do you have access to "related" folder in my project folder with PDFs of papers, related papers?

Worked for 22s >

Yes—I can access related-paper PDFs synced to this project under **sources**, including *Writing is thinking*, *Mapping human and AI contributions across stages of academic writing Using SLICE*, and *Who is responsible when AI helps to write science*.

I don't see a folder literally named **"related"** in this project mirror. If you mean a separate folder in your OneDrive project directory, its contents have not yet been checked.

I want you to write the related work section, section 2.1. I paste below the related work that we wrote from my previous paper, okay? You should have its Word file, that we have compared this paper, and also I attached many, many PDF files of related papers. So give me a concise 2.1 section.-----

Worked for 1m 23s >

References for integration into your bibliography:

- Porsdam Mann, S., Vazirani, A. A., Aboy, M., Earp, B. D., Minssen, T., Cohen, I. G., & Savulescu, J. (2024). Guidelines for ethical use and acknowledgement of large language models in academic writing. *Nature Machine Intelligence*, 6, 1272–1274. [@https://doi.org/10.1038/s42256-024-00922-7](https://doi.org/10.1038/s42256-024-00922-7)
- Getto, G., Kelley, S., & Vance, B. (2026). How to write with GenAI: A framework for using generative AI to automate writing tasks in technical communication. *Journal of Technical Writing and Communication*, 56(2), 111–144. [@https://doi.org/10.1177/00472816251332208](https://doi.org/10.1177/00472816251332208)
- Nature Reviews Bioengineering. (2025). Writing is thinking. *Nature Reviews Bioengineering*, 3, 431. [@https://doi.org/10.1038/s44222-025-00323-4](https://doi.org/10.1038/s44222-025-00323-4)
- Hurshman, C., Porsdam Mann, S., Savulescu, J., & Earp, B. D. (2025). *Authorship without writing: Large language models and the "senior author" analogy*. Preprint, supplied version.
- Earp, B. D., Guernon, A.-S., & Porsdam Mann, S. (2026). *A substantial human contribution: Do we deserve to be authors of this essay?* Preprint, supplied version. [@https://www.researchgate.net/publication/403018576](https://www.researchgate.net/publication/403018576)
- Van Woudenberg, R., Ranalli, C., & Bracker, D. (2024). Authorship and ChatGPT: A conservative view. *Philosophy & Technology*, 37, Article 34. [@https://doi.org/10.1007/s13347-024-00715-1](https://doi.org/10.1007/s13347-024-00715-1)

- Earp, B. D., & Porsdam Mann, S. (2026). *Mapping human and AI contributions across stages of academic writing using SLICE—a preliminary LLM-based method*. Preprint, supplied version. <https://www.researchgate.net/publication/406991607>

## 2.1 is RELATED WORK ??

Yes—use **2.1 Related work**. The text already serves that purpose; the broader heading accommodates guidance, expertise, authorship, and contribution mapping.

give 1 Introduction

Logs can provide rich amount of data

## Data mining

The amount of prose generated by AI should not be equated with the amount of intellectual agency retained by the human researcher. A researcher may have AI draft most of the prose yet exercise high intellectual agency by designing the argument, scrutinizing substantive claims, rejecting alternatives, supplying evidence, and repeatedly redirecting AI. Conversely, a researcher may generate substantial portions of the final prose while relying on AI for higher-level decisions about framing, argument, structure, or interpretation. The relevant question is therefore not simply who wrote the words, but who exercised intellectual agency over the development of the scholarly work.

The relevant question is not simply who wrote the words, but who exercised intellectual agency over the development of the scholarly work. ... ----- and some refs in it

RE: four papers developed by the same researcher with AI assistance. The cases encompass software engineering, higher education, and the study of human-AI co-writing.--- You have to clarify that there was no bias because the papers were written before the idea of this entire study was actually initialized. So they were developed. So the researcher had done those AI collaboration and co-writing in a very free way without any plan to even use those logs later.

We can clarify that the interactions preceded the study and were not shaped by its coding framework. This addresses **reactivity**, but does not eliminate retrospective selection or interpretation bias.

**Abstract:** 150 words

**Keywords:** Human-AI co-writing; academic writing; generative AI; human-AI collaboration; technical writing

update m,y Disclosure of AI Use

The relative contributions of the human researcher and AI to the different research activities involved in producing this paper are reported in Table 1. These ratings provide the author's retrospective assessment of the human–AI division of contribution in this specific study. GPT-5.6 Sol (OpenAI) was used as an AI writing assistant during the development of the manuscript, including support for structuring, drafting, revising, and refining the text. The human author retained responsibility for the research direction, evaluation of AI outputs, verification of content, and final decisions regarding the manuscript. The complete log of human–AI interactions underlying the development of this paper is available in the accompanying open data repository, enabling examination of the co-writing process described in this study.

[https://chatgpt.com/s/cx\\_6ac42b1335c081918bb7da26ff504a4b](https://chatgpt.com/s/cx_6ac42b1335c081918bb7da26ff504a4b)